Vocabulary resonance in US trademark filings:
A measurement construct for commercial novelty, with validation and applications

Eric Silver*

June 2026 working paper

Abstract

The DeDeo prospective/retrospective Kullback–Leibler resonance framework (Barron et al., 2018; Murdock et al., 2017) was developed on cognitive corpora in which the author of each document is the agent whose novelty is being measured: Darwin's reading notebooks, parliamentary speeches of the French Revolution. This paper applies the framework to a qualitatively different corpus type, the goods/services descriptions of US trademark filings, and introduces the resulting construct, denoted ΔKL, as a firm-level measure of commercial novelty that is cheap to compute, available for millions of firms that never patent, and empirically distinct from patent-track invention.

Three validation results anchor the construct. First, ΔKL predicts commercial outcomes in a consistent forward-favoring direction: 5-year mark renewal conditional on completed registration (innovator-tail vocabulary renews at $\sim$87% versus $\sim$80% for the going-out-of-fashion tail in the diagnostic class), gross margin on an SEC-matched panel (+3.2pp per σ, $t = 5.3$), excess stock returns at 1–4 year horizons (+1.5 to +5.0pp per σ, decaying by year five), and eventual public listing among first-time filers. Notably, prospective surprise alone predicts none of these; only the signed resonance does. Second, ΔKL is orthogonal-to-negative with patent counts within firm (-0.048σ, $t = -7.9$ on 14,463 matched firms), while expert "most innovative" lists sit $+0.34\sigma$ above the panel mean ($p < 0.001$); the construct captures a market-facing dimension of novelty that patents miss. Third, the construct supports direct measurement of cross-industry idea diffusion: software- and tech-services-origin themes arrive in transport and advertising/retail with 1–13 year lags, theme adoption follows Bass curves with median $q/p \approx 7$, and the cross-class flow is strongly asymmetric (33 outflow edges from software/tech, 4 returning).

The paper closes with a research agenda the construct opens: whether ideas that transfer between industries hold their value, whether suppliers to a new theme outperform its deployers, and how demographic concentration within industries relates to where new themes originate. Code and firm-year panels are public at `https://github.com/ericsilver/novelty`.

1 Introduction

Innovation measurement leans heavily on patents. Patent-based measures are well-validated for technical invention but cover a narrow slice of commercial novelty: most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer a complementary window. Nearly every commercially serious

*Independent researcher; formerly a PhD candidate at Carnegie Mellon University. Comments to `epsilver@gmail.com`. The author's current employment is unrelated to this work, and the views expressed are the author's alone.

product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year. The text of those descriptions, however, has been largely unused; the literature has worked with filing counts and filing metadata rather than filing language.

Murdock et al. (2017) introduced a measurement framework for directional novelty in time-ordered text corpora: for each document, compare its token distribution to the aggregated distribution of past documents (prospective surprise) and future documents (retrospective surprise). The signed difference of those two quantities they call *resonance*: it is positive when a document is unusual against its recent past and close to its near future, negative for the inverse. The framework was developed on Charles Darwin's reading notebooks and extended in Barron et al. (2018) to French Revolution parliamentary debates. Both corpora share a salient property: the text is authored by the agent whose novelty is the object of measurement.

This paper applies the same apparatus to trademark goods/services text, which is commercially incentivised, typically drafted by counsel, and exists to define legal protection scope rather than to express cognition. Whether the resonance framework produces interpretable structure on text of this kind is the threshold question, and the answer this paper documents is yes. The construct, denoted ΔKL, behaves as a forward-position measure: filings whose vocabulary leans toward where their industry's language is going belong to marks that are renewed more often, firms with higher margins and better stock returns, and first-time filers who more often eventually go public. The same signal is uncorrelated-to-negatively correlated with patenting within firm, while firms on expert "most innovative" lists score high on it; ΔKL captures a market-facing dimension of novelty that the patent record does not.

Because the construct is computed from public text with public code, it scales to questions that firm-survey or patent-linked measures cannot easily reach. The paper demonstrates one such application in depth, measuring the diffusion of business vocabulary across industries directly at the phrase and theme levels, and sketches a research agenda: the portability of transferred ideas, the relative fortunes of suppliers versus deployers of a new theme, and the relationship between demographic concentration within an industry and where new themes are born.

Section 2 defines the construct, the corpus, and known fragilities. Section 3 reports the commercial-outcome validation. Section 4 reports the within-firm patent contrast and the expert-list discriminant. Section 5 reports the cross-industry diffusion application. Section 6 reports a base-rate discipline result on entrant-vs-incumbent classification. Section 7 states the research agenda. Section 8 bounds the claims. Section 9 is reproducibility.

2 The construct

2.1 The DeDeo apparatus on trademark text

For a time-ordered sequence of documents, with each document at year t producing a token distribution P_t, the framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{past}})$, where Q_{past} is the aggregate token distribution of documents in years $t-W$ to $t-1$.
- *Retrospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{future}})$, where Q_{future} is the aggregate distribution of documents in years $t+1$ to $t+W$.

The signed resonance is

$$\Delta\mathrm{KL}_t \;=\; \mathrm{KL}(P_t \,\|\, Q_{\mathrm{past}}) - \mathrm{KL}(P_t \,\|\, Q_{\mathrm{future}}).$$

Positive ΔKL marks a document whose vocabulary is unusual relative to its recent past and close to its near future: the field moved toward this filing. Negative ΔKL marks the inverse, language that looked normal at filing and looked played-out in hindsight: the field moved away.

2.2 Corpus and conditioning

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work in this paper uses the complete software/electronics class (NICE 009; 1.81M filings), and the diffusion application uses four classes selected to exercise canonical cross-industry flow: software (009), scientific and technical services (042), transport (039), and advertising and retail (035), totalling 2.32 million granted filings 1990–2024. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees.

Two conditioning rules apply throughout. The corpus is restricted to *granted* filings, screening out speculative or defensive applications that did not survive examination. And wherever the paper reports a mark-renewal rate, the denominator is further restricted to filings that reached registration: the decision to register and the decision to renew are different commercial acts with different drivers, and pooling them conflates examination outcomes with maintenance outcomes.

2.3 Computation

For each granted filing at year t in class c with goods/services text producing token distribution P (unigram + bigram, after a standard analyzer that drops stopwords), the reference distributions Q_{past} and Q_{future} are computed from all granted filings in class c in the respective windows. I use $W = 5$ years (flat window) with Dirichlet smoothing ($\alpha = 1$ per cell). The within-class vocabulary is the set of unigrams and bigrams with document frequency ≥ 50.

The signal depends on having enough past and future filings to populate the references. Figure 1 shows the reference-window density by year for two diagnostic classes; the 5-year past window is incomplete before 1990 and the 5-year future window is incomplete after 2020, and both zones are excluded from the construct's interpretable range. Within the supported window the classes analyzed here have between $\sim$3,500 and $\sim$80,000 filings per year.

2.4 The construct at the case level

Figure 2 grounds the construct in recognizable examples. The two scatter axes are prospective surprise and retrospective surprise, with ΔKL as the diagonal residual. Filings below the diagonal are *innovator*: the language looked unusual at filing and became standard afterwards. Filings above the diagonal are *tired*: the past resembles the filing more than the future does.

2.5 Known fragilities

The construct has documented fragilities on this kind of text. On the complete Class 009 corpus, prospective and retrospective KL are correlated at $r = +0.971$ across 1.56M scored filings; signed ΔKL is therefore a small residual of two large, highly correlated quantities, and the residual is noisy at the per-filing level. The magnitude $|\Delta\mathrm{KL}|$ inflates monotonically for short filings (0.293

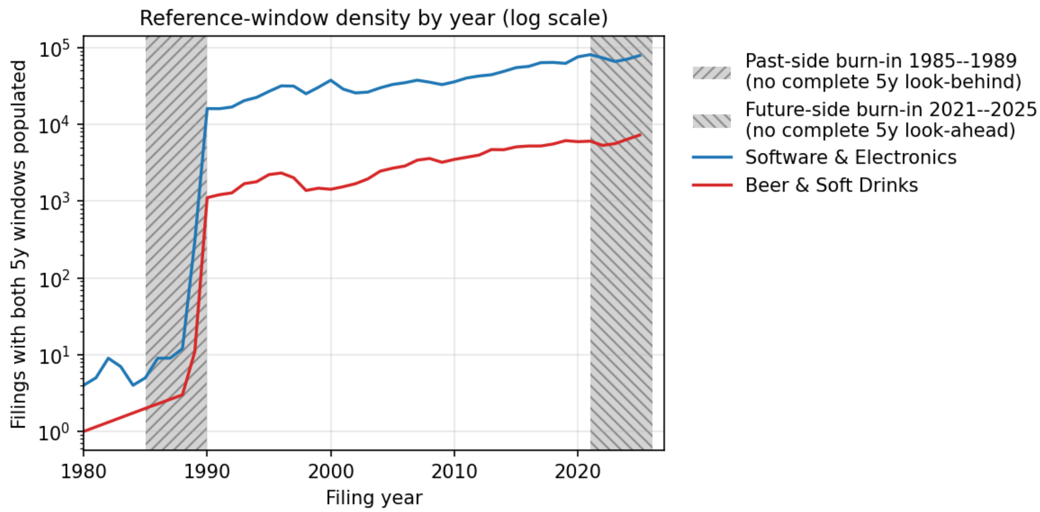

Figure 1: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

for documents under 5 terms, decaying to 0.147 for documents of 40 or more terms), a length artifact that requires control in any per-filing analysis. Pre-1995 readings are contaminated by thin past references. Window sensitivity is bounded: across $W \in \{3, 5, 7\}$ years, 6.9–10.4% of filings flip sign, well under conventional fragility thresholds. The per-filing noise attenuates at every aggregation used in this paper: firm-year means, token-level pools, and theme-level prevalence trajectories.

3 Validation against commercial outcomes

If ΔKL measures something commercially real, it should predict outcomes that cost firms money: whether a registered mark is maintained, whether the firm earns margins, whether early vocabulary position precedes growth. It does, and in a consistent direction. Forward-leaning vocabulary is good news at every level of measurement.

3.1 Registration, renewal, and listing among first-time filers

Figure 3 traces a debut filing (the owner's first-ever filing) through three successive outcomes: whether the application reaches registration, whether the registered mark is still alive and maintained five years on, and whether the owner ever appears in the SEC EDGAR listed universe. The panel sample is 1.49 million debut filings in the four-class build, of which 715,904 reached registration; the second and third outcomes are computed on the registered subset only.

The registration step (panel A) is the largest filter, and it selects on the KL *level*: applications with higher prospective or retrospective surprise register more often (rising from $\sim$38% to $\sim$57% across the distribution), consistent with the examination standard rewarding distinctiveness. Conditional on registration (panel B), renewal is forward-favoring in the signed ΔKL: marks at the innovator tail renew at $\sim$93% against $\sim$84% at the going-out-of-fashion tail, while the directional pieces show only a shallow dip-and-recover. The same ordering holds on the complete software/electronics class without the debut restriction (innovator tail $\sim$87%, flux-neutral middle $\sim$82%, negative tail $\sim$80%).

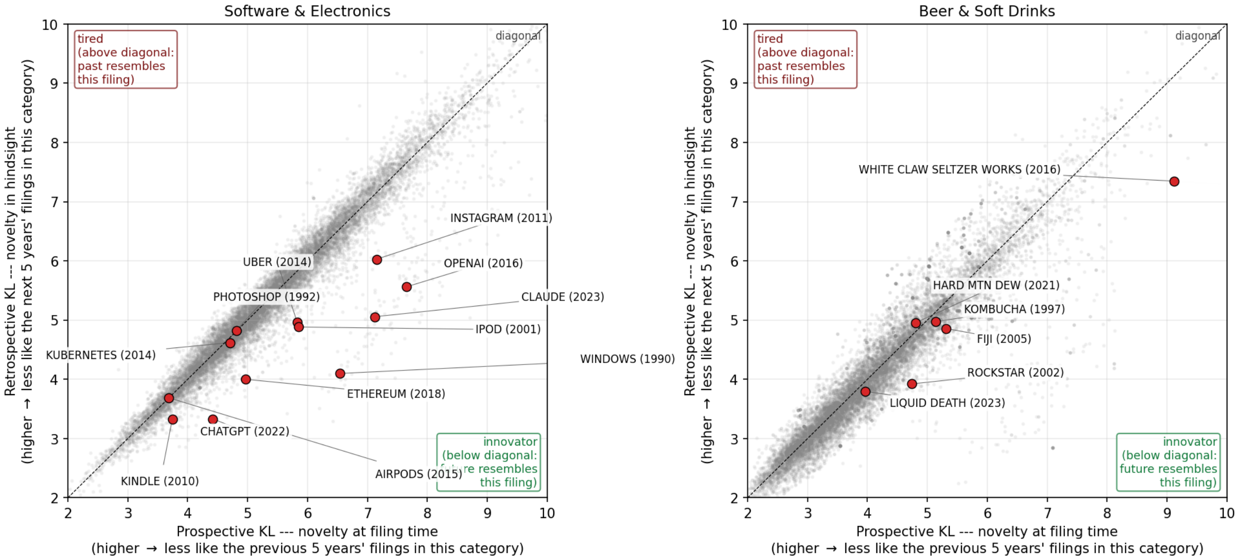

Figure 2: Prospective KL versus retrospective KL with ΔKL as the above-diagonal/below-diagonal residual. Labeled examples are recognizable brand introductions. *Left:* software/electronics. The innovator quadrant contains Kindle, Photoshop, Kubernetes, Instagram, Uber, ChatGPT, OpenAI, and iPod; Windows (1990) sits on the tired side. *Right:* a beer/soft-drinks diagnostic class, with White Claw and Liquid Death on the innovator side.

The token-level pattern matches. Conditional renewal rates for marks containing example tokens: positive-tail tokens renew highest (*blockchain* 99.7%, *cryptocurrency* 99.9%, *cbd* 100%), flux-neutral middle tokens in between (*organisational* 92%, *conformity* 85%, *freighting* 95%), and negative-tail tokens lowest (*typesetting* 63%, *annuity* 82%, *underwriting* 84%). The conditioning matters: middle-vocabulary applications register at ~15–26% while tail-vocabulary applications register at ~30–60%, so an unconditional survival statistic on these tokens is dominated by the registration-step gap rather than the renewal-step gap.

Among registered debut filings, the probability that the owner ever appears in the SEC EDGAR listed universe (panel C) also rises in ΔKL, from ~0.40% at the negative tail to ~0.55% at the positive tail, against a 0.47% base rate. The debut restriction matters here: at the whole-firm level, selection into public markets is dominated by filing volume (a tenfold increase in filings raises the odds of listing roughly fivefold), and firm-level ΔKL quintiles show no lift. The construct's signal for eventual public listing is concentrated at the moment of entry, before portfolio-scale effects swamp it.

3.2 Margins and returns on the SEC panel

On a panel of 5,319 firm-years (1,597 SEC-matched trademark owners, 2009–2024), trailing 3-year mean ΔKL predicts gross margin at +3.2pp per σ ($t = 5.3$), controlling log revenue and R&D intensity. The decomposition is instructive: prospective surprise enters at +14.6pp and retrospective surprise at −19.1pp (both $|t| > 3.5$), so the margin premium attaches to vocabulary that is unusual against the past *and* close to the future, exactly the resonance configuration. On first-time filers alone, where reverse causality from established success is weakest, the effect remains at +1.7pp per σ ($n = 6{,}525$ firms).

Excess stock returns over the S&P 500 show the same signature. Firm-year mean ΔKL predicts

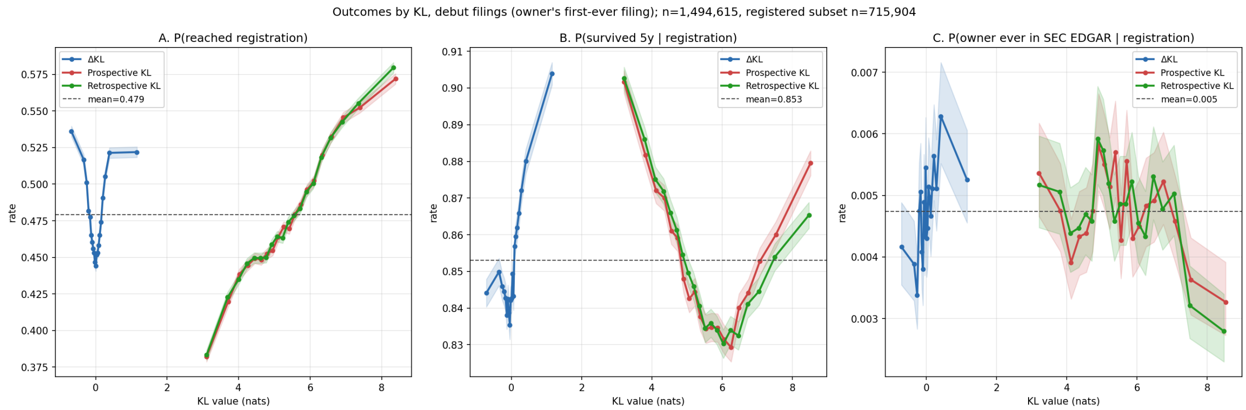

Figure 3: Outcomes for debut filings ($n = 1{,}494{,}615$; registered subset $n = 715{,}904$) by KL value, in quantile bins with 95% intervals. *A:* P(reached registration). *B:* P(mark alive and maintained at 5 years | registration). *C:* P(owner ever in SEC EDGAR | registration).

+1.5pp per σ at the 1-year horizon ($t = 3.0$), +3.5pp at 2 years, +5.0pp at 3 years, +4.1pp at 4 years, and decays to insignificance at 5 years. Prospective surprise alone predicts approximately zero at every horizon. Mere novelty at filing time carries no return information; novelty the future moves toward does. A larger headline figure from a debut-only specification (+28pp at 4 years) does not survive a sample-size diagnostic and is not claimed; the firm-year profile is the defensible one.

4 Distinctness from patents

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not.

The within-firm analysis uses a published panel of 174,569 firm-years covering 14,463 trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024); the panel and matching code are in the public repository. The matching normalizes both sides (strip corporate-form suffixes and punctuation, lowercase) and joins on exact normalized-name equality. I fit a two-way (firm + year) fixed-effects regression of standardized firm-year mean ΔKL on $\log(1 + \text{patents})$ with firm-clustered standard errors.

Table 1: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1 + \text{patents})$.

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling log(n_filings)	−0.051	0.006	−8.4
Lead/lag: patents_{t-1}	−0.054	0.009	−5.8
patents_{t}	−0.019	0.010	−1.9
patents_{t+1}	+0.010	0.010	+1.0

The coefficient is negative and precisely estimated, strengthening under filing-count control.

The strongest negative coefficient sits on patents$_{t-1}$: in years after a firm patents heavily, its trademark vocabulary regresses toward class-typical. A reading consistent with the data is temporal substitution between patent prosecution and market-facing brand work; either way, the two measures are not interchangeable.

The contrast with expert judgment completes the discriminant picture. Across the 7,431-firm public panel, firms appearing on the BCG Most Innovative lists (2021–23) sit $+0.28\sigma$ above the panel mean in ΔKL ($n = 31$ matched, $p = 0.012$), firms on the MIT Technology Review 50 Smartest Companies list (2015) sit $+0.50\sigma$ above ($n = 17$, $p = 0.0006$), and the pooled union sits $+0.34\sigma$ above ($n = 43$, $p = 0.0003$). Patent counts are uncorrelated with ΔKL on the same panel (Pearson +0.010). The expert lists reward something patents do not capture, and ΔKL tracks it. The matched samples are small and the result is a discriminant check, not a headline.

5 Application: cross-industry diffusion, measured directly

Because ΔKL is computed from open text, the same corpus supports measuring where business vocabulary is born and how it travels. This section demonstrates the construct's reach beyond firm-level validation.

5.1 Phrase-level transit

For 14 curated business-model phrases, I record the first year each of the four classes crosses a 1% prevalence floor among granted filings.

Table 2: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size (transport has roughly one-eighth the volume of software), and the asymmetry is a four-class result that a 45-class build would test properly; the current evidence is consistent with the directional reading but cannot exclude a corpus-construction alternative.

5.2 Theme-level structure

A theme-level analysis with unsupervised topic extraction corroborates the phrase-level pattern without depending on which phrases were picked. I fit LDA with $T = 50$ on a stratified 200,000-filing sample (vocabulary $V = 69{,}361$) and transform across the full 2.32M-filing pool. The resulting topics are recognizable business-model components (health/wellness; autonomous loading and material handling; retail store services; electronic transmission and computer networks; vehicles and traffic).

For each (theme, class) cell that reaches the 1% floor and is tracked at least six years, I fit the Bass diffusion model (Bass, 1969). Of 118 retained fits ($R^2 \geq 0.7$), the median innovation coefficient p is 0.017, the median imitation coefficient q is 0.114, and median $q/p \approx 7$, in the range typically reported for consumer-product adoption. Most fits are platform-like; a small tail is fad-shaped (renewable-energy and virtual-reality themes show $q/p > 40$, with steep adoption and visible recession).

Counting themes that reach at least two classes and identifying the origin class produces a directed flow graph: of 33 multi-class themes, 33 outflow edges originate in software/tech-services and only 4 return there. The same asymmetry reproduces under independently-fitted NMF on the same sample and vocabulary, with seven of ten curated seed phrases recovering a clear best topic. Both methods are bag-of-words factorizations, so this robustness is within-paradigm; a sentence-embedding or graph-community extraction is the natural cross-paradigm test and is not run here.

6 Do new themes come from incumbents or entrants?

The corpus invites population-share questions, and it punishes them when the base rate is ignored. As a worked example with substantive interest of its own: when a new theme appears, is it carried by new entrants or by incumbents diversifying in? Schumpeter (1942) introduced the distinction (Mark I entrant-led, Mark II incumbent-led), and the standard empirical form reports the entrant share among early adopters.

Applying the standard form to the 50 LDA themes, the 50 earliest carriers per theme are 73.7% entrants on average, and 48 of 50 themes are majority-entrant. The raw numbers look like strong entrant-led adoption. But the corpus's debut rate among all granted filings in this period is 76.4% under year-matched weighting; most filings are by owners debuting that year. Against that baseline the theme-carrier excess is −2.7pp, and 28 of 50 themes sit below baseline. Themes inherit the corpus's high debut rate without adding to it.

The residual pattern is the interesting part. The themes carried disproportionately by incumbents relative to baseline include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing. The canonically new technology themes of the 2010s and 2020s lean incumbent: established firms diversifying into AI and cloud pull the vocabulary signal forward at least as much as de-novo entrants do. This is consistent with the within-firm patent contrast of Section 4; the firms moving the vocabulary are not necessarily the firms moving the patents.

7 A research agenda the construct opens

Each of the following is computable from the public corpus and panels with the apparatus of this paper. None is run to completion here.

Are transferred ideas as valuable away from home? The flow graph identifies, for every theme, its origin class and arrival classes. Linking carrier filings to the SEC panel tests whether a theme earns the same margin and survival premia in destination industries as in its origin. A first-pass within-firm estimate finds a contemporaneous gross-margin lift of +1.4pp per σ of borrowed novelty ($t = 2.1$) that fades within a year; whether portability is the rule or the exception is open.

Suppliers versus deployers. When a phrase crosses industries, the carrier filings split into firms whose offer *is* the new thing and firms whose offer *supplies* the firms doing the new thing, the sellers of pick-axes rather than the panners for gold. The goods/services text is rich enough to separate the two roles, and the gold-rush intuition that suppliers outperform deployers is testable against renewal, margins, and listing.

Demographic concentration and where themes are born. Surname-imputed demographic composition of filing owners, applied to US-domiciled individual filers, identifies industry enclaves in which a demographic group is heavily over-represented. Preliminary work finds enclave filings sit on the innovation tail of ΔKL rather than the harvest tail, and that owners who diversify across classes list publicly at many times the single-class rate. Whether enclave industries disproportionately originate themes, and whether enclave-rooted firms carry them outward, is a direct join of the demographic panel to the flow graph.

External codification. The USPTO ID Manual periodically standardizes acceptable goods/services wording. If high-ΔKL filings use terms the Manual later codifies, while those terms are still rare in class, the construct anticipates institutional recognition, the strongest available external validation and one that requires only the Manual's version history.

Cross-paradigm theme extraction. The diffusion structure has been shown method-stable within bag-of-words factorizations. A sentence-embedding clustering or graph-community extraction on the same corpus would establish (or refute) paradigm independence.

8 Scope and limits

The paper makes four substantive claims. ΔKL, computed as in Barron et al. (2018); Murdock et al. (2017) on granted USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. It predicts commercial outcomes in a consistent forward-favoring direction: renewal conditional on registration, gross margin, 1–4 year excess returns, and eventual listing among first-time filers, with the signed resonance carrying the signal and prospective surprise alone carrying none. It is empirically distinct from patent-track invention within firm while aligning with expert innovation judgment. And it supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and strongly asymmetric flow out of software and tech services.

The paper rejects two readings the construct does not support. ΔKL does not measure innovation in any strong sense; the defended primitive is lexical resonance on commercial text, and the validation evidence bounds the interpretation. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic.

The renewal results are conditional on completed registration and do not describe the full application distribution; an applicant whose vocabulary keeps the filing from registering is excluded from the renewal denominator. The margin and returns results are associational, identified within an SEC-matched panel that itself selects on scale. The diffusion asymmetry is a four-class result pending a 45-class build. The expert-list discriminant rests on 43 matched firms.

Questions owed to follow-up work: the four agenda items of Section 7; whether the conditional renewal pattern extends beyond the diagnostic class to the full corpus pooled; and whether the residual renewal gap between forward and backward tails reflects vocabulary position itself or selection on owner covariates.

9 Reproducibility

Code, firm-year panels, and analysis outputs are available at `https://github.com/ericsilver/novelty`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the underlying TRTYRAP backfile is approximately 12 GB streamed once). The published panels include the SEC-matched firm-year ΔKL panel ($n = 59{,}033$) and the PatentsView-joined panel ($n = 174{,}569$). Analysis scripts and their outputs are indexed in the repository `README.md`.

References

Barron, A. T., Huang, J., Spang, R. L., & DeDeo, S. (2018). Individuals, institutions, and innovation in the debates of the French Revolution. *Proceedings of the National Academy of Sciences*, 115(18), 4607–4612.

Bass, F. M. (1969). A new product growth for model consumer durables. *Management Science*, 15(5), 215–227.

Dinlersoz, E., Goldschlag, N., Myers, A., & Zolas, N. (2018). An anatomy of US firms seeking trademark registration. *NBER Working Paper* No. 25038.

Graham, S. J. H., Hancock, G., Marco, A. C., & Myers, A. (2013). The USPTO trademark case files dataset: Descriptions, lessons, and insights. *Journal of Economics & Management Strategy*, 22(4), 669–705.

Hoberg, G., & Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, 124(5), 1423–1465.

Murdock, J., Allen, C., & DeDeo, S. (2017). Exploration and exploitation of Victorian science in Darwin's reading notebooks. *Cognition*, 159, 117–126.

PatentsView. (2024). Patent and assignee data, US Patent and Trademark Office. `https://patentsview.org`.

Schumpeter, J. A. (1942). *Capitalism, Socialism and Democracy*. Harper & Brothers.